

POLI110 Tutorial 5: Teaching Assistant Notes & Measurement Error

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Key Concepts

1. Validity

Definition: The extent to which a variable/measure corresponds to the concept it is supposed to measure.

Requirements for validity:

- Should capture the concept as fully as possible
- Should not capture other concepts
- Should maintain comparability across units

2. Measurement Error

Let X_{True} be the true value and $X_{Observed}$ be the measured value. The measurement error (ϵ) is defined as:

$$\epsilon = X_{True} - X_{Observed}$$

A. Random Measurement Error

- Error has mean zero: $E(\epsilon) = 0$
- Symmetric distribution around true value
- No systematic pattern in errors



Figure 1: Random Measurement Error: Note the symmetric distribution around zero ($X_{True} - X_{Observed} = 0$)

B. Systematic Measurement Error (Bias)

- Error has non-zero mean: $E(\epsilon) \neq 0$
- Consistent direction of error
- Can be positive (overestimation) or negative (underestimation)

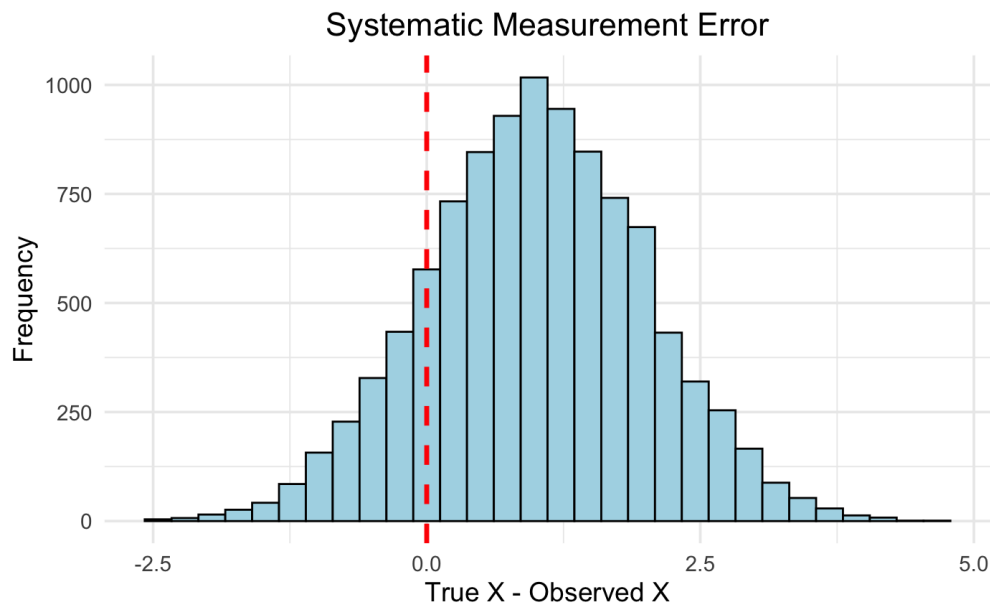


Figure 2: Systematic Measurement Error: Distribution shifted away from zero

C. No Measurement Error

- Perfect measurement: $X_{True} = X_{Observed}$
- Therefore, $\epsilon = 0$ for all measurements
- Rare in practice but ideal scenario

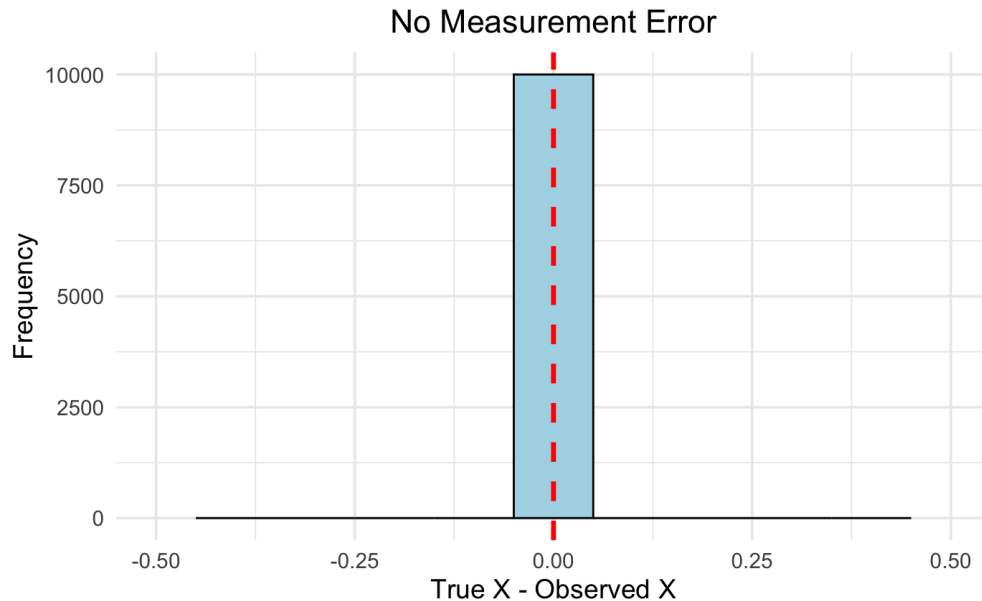


Figure 3: No Measurement Error: All measurements exactly at true value

Examples of Measurement Error in Political Science

Electoral Participation Studies

Consider a research project measuring voter turnout in a national election. This example demonstrates how different types of measurement error can manifest in a single study:

A national post-election survey (n=2000) reported 78% turnout, while official records showed 62% actual turnout. This discrepancy illustrates both random and systematic measurement errors.

The systematic error component ($E(\epsilon) \neq 0$) emerges from social desirability bias, where non-voters consistently report voting to appear as responsible citizens. This creates a positive bias, with $X_{Observed}$ systematically exceeding X_{True} .

Random error appears through:

- Memory errors in recall ($\pm 2-3\%$)
- Survey response recording variations
- Sample coverage fluctuations

Measuring Democratic Quality

The measurement of democratic quality across nations presents complex measurement challenges:

When measuring democracy using a 0-10 scale, established democracies often receive scores of 9-10, potentially masking meaningful variations in democratic quality.

This ceiling effect represents systematic measurement error through:

$$X_{Observed} = \min(X_{True}, 10)$$

The truncation at the upper bound creates systematic underestimation of differences among high-performing democracies. This measurement limitation affects comparative analyses and time-series studies of democratic development.

Political Trust Measurement

Measuring political trust illustrates the challenge of validity in concept operationalization:

$$Trust_{Observed} = Trust_{True} + \epsilon_{systematic} + \epsilon_{random}$$

Where:

- $\epsilon_{systematic}$ reflects institutional context bias
- ϵ_{random} captures individual response variations

Cross-national studies show systematic variation in how citizens interpret trust questions, with some cultures consistently reporting lower trust levels despite similar institutional performance.

This systematic error component creates challenges for cross-national comparisons and requires careful methodological adjustments, such as:

1. Cultural calibration of survey instruments
2. Multi-method measurement approaches
3. Contextual variable controls

Implications for Research Design

These examples illustrate the importance of:

1. Identifying potential sources of both random and systematic error during research design
2. Implementing appropriate methodological controls
3. Acknowledging measurement limitations in analysis and conclusions

Researchers must balance measurement precision with practical constraints, often employing multiple measurement strategies to triangulate true values and minimize both types of error.

Class Exercise

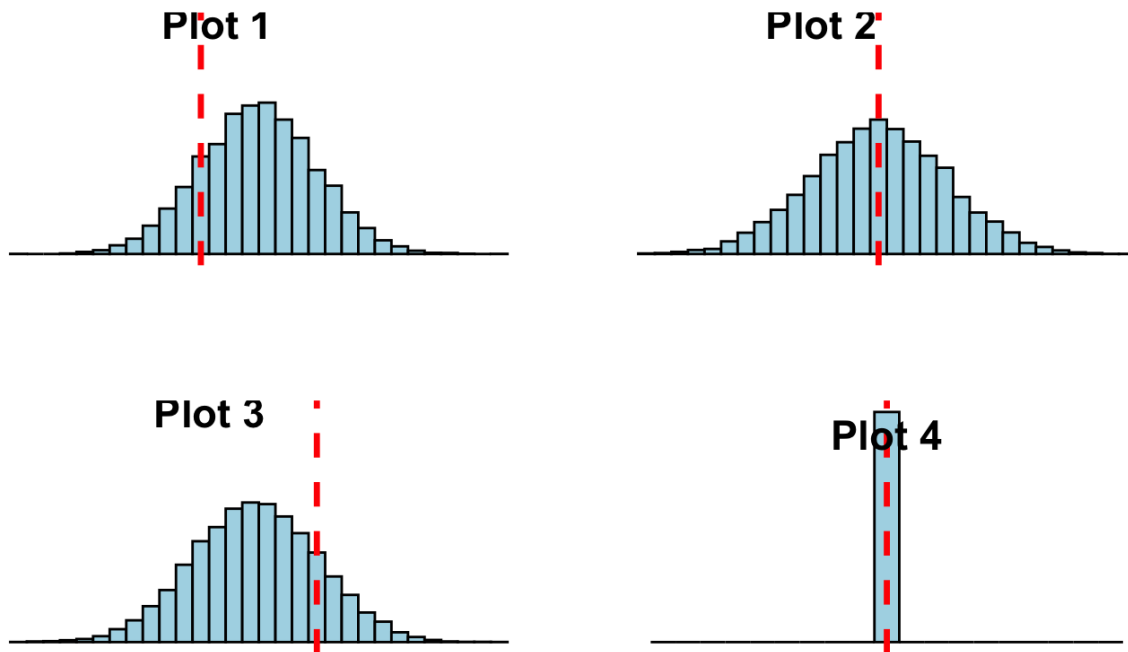


Figure 4: Different Types of Measurement Error (X1-X4)

Analysis Task:

1. For each measure (X1-X4), determine:
 - The type of measurement error present
 - Whether $E(\epsilon) = 0$ or $E(\epsilon) \neq 0$
 - Direction of bias (if any)
2. Support your answers by:
 - Analyzing the position of distribution relative to $X_{True} - X_{Observed} = 0$
 - Examining the shape and spread of the distribution
 - Considering the mathematical properties of each error type

Answer Key

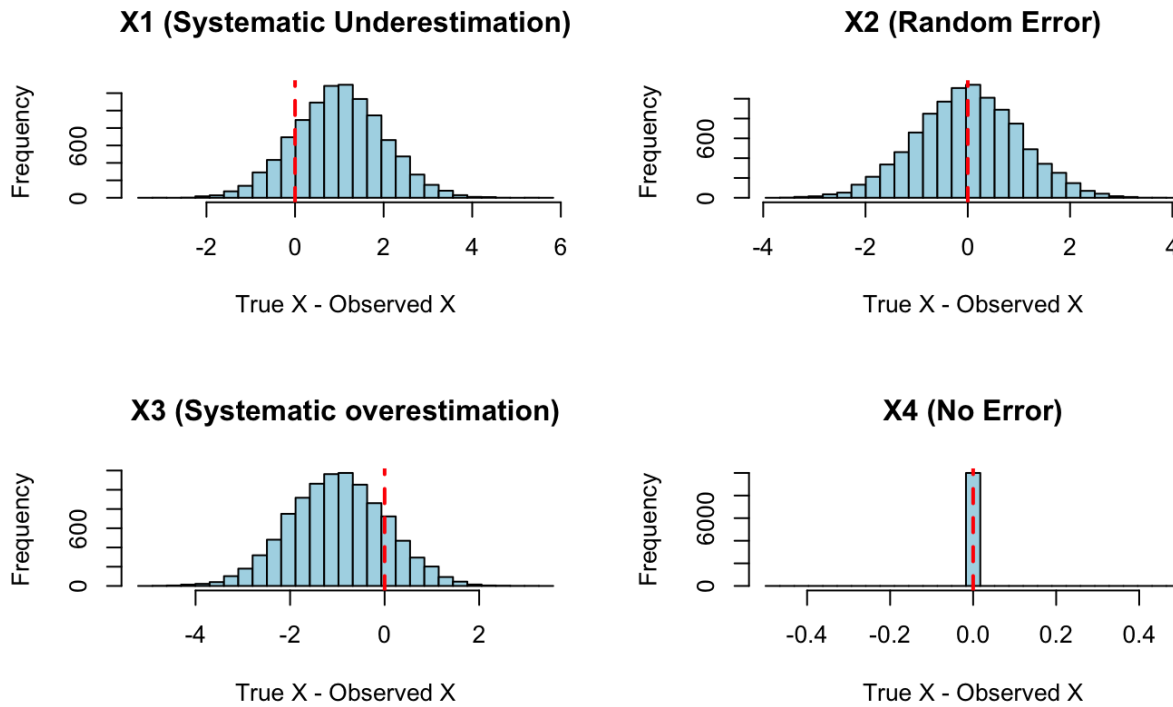


Figure 5: Different Types of Measurement Error

Key Features for Analysis:

1. X1 (Systematic Overestimation):
 - Distribution centered left of red line ($X_{True} - X_{Observed} > 0$)
 - Consistently overestimates true value
 - $E(\epsilon) > 0$
2. X2 (Random Error):
 - Symmetric around red line
 - Equal probability of over/under-estimation
 - $E(\epsilon) \approx 0$
3. X3 (Systematic Underestimation):
 - Distribution centered right of red line ($X_{True} - X_{Observed} < 0$)
 - Consistently underestimates true value
 - $E(\epsilon) < 0$
4. X4 (No Error):

- Single spike at zero
- Perfect measurement
- $\epsilon = 0$ for all measurements